

# A New Approach to the Image Resizing Using Interpolating Rational-Linear Splines by Continued Fractions<sup>\*</sup>

Min Hu<sup>a,\*</sup>, Jieqing Tan<sup>b</sup>, Feng Xue<sup>a</sup>

<sup>a</sup>*College of Computer and Information Science, Hefei University of Technology, Hefei 230009, China*

<sup>b</sup>*Institute of Applied Mathematics, Hefei University of Technology, Hefei 230009, China*

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## Abstract

This paper proposes a new image resizing method based on splines, which is constructed by the Thiele-type continued fractions and the Newton interpolation. Like most interpolation techniques proposed in the past, this paper focuses on determining pixel values using the relationship between neighboring points. The interpolation is carried out by choosing the osculatory interpolants in every subinterval effectively. The small image details and step edges are preserved based on the proposed interpolation method. The simulations indicate that this method achieves better image results than those by conventional methods in terms of error, PSNR and image quality performance.

*Keywords:* Image resizing; Newton interpolation; Thiele-type continued fractions; Rational spline

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## 1 Introduction

Image resizing is an essential task in many image processing applications such as satellite imaging, medical imaging, digital photography, multimedia etc. Over-sampling facilitates image interpretation, better visualization, etc., whereas sub-sampling facilitates fitting of large images into smaller display area, creating icons etc. Zooming of images can be accomplished with interpolation ([1]–[7]) where the procedure is to fit the image data with a continuous model and then re-sample the function on the grid appropriate for the desired scaling. The nearest neighbor interpolation, which uses a polynomial spline of order zero ( $n = 0$ ), requires only one point and results in blocky images. Bilinear interpolation with  $n = 1$  requires two points and produces better images, but tends to blurry image details. To counter the blurring and jaggedness problems, interpolators

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<sup>\*</sup>Corresponding author.

Email address: [jsjxhumin@hfut.edu.cn](mailto:jsjxhumin@hfut.edu.cn) (Min Hu).

using more neighboring points are proposed in many literature. For example, cubic splines are used to reduce blurring by increasing the smoothness of interpolator ([9]).

As we know, the image data of two-dimensional plane have such characters that pixel gray scale is influenced by material of the object, the nature of the background object, intensity and angle of the light. Thus the image's pixel data are neither purely linear nor purely nonlinear. In these cases, the method of rational interpolation plays an irreplaceable role. Rational functions have been extensively used for image filtering and restoration, image enhancement as well as image interpolation ([5, 6]) because a rational function is a universal approximator (it can approximate any continuous function arbitrarily well), and possesses better extrapolation capabilities compared with a polynomial function. Since rational function operators are able to adapt their behavior with respect to the local source content, they exhibit remarkable performance in the previously mentioned applications. For this reason, this paper constructs rational splines by using the Thiele-type interpolation theory ([8]) and polynomial splines by using the Newton's interpolation theory ([10]). These methods regenerate an image of improved quality by inserting the missing high frequency components using the gradient that provides the relationship between neighboring points. The paper is organized as follows. In Section 2 spline functions based on the Newton interpolation is introduced and Thiele-type interpolation is presented. In Section 3, the proposed interpolation method as a combination of a linear polynomial spline and a rational spline is described. In Section 4, results of image resizing are presented in comparison with other conventional approaches. Conclusions are presented in Section 5.

## 2 Newton Interpolation and Thiele-type Interpolation

Suppose  $f(x)$  is a function defined on a subset  $G$  of the complex plane and  $X = \{x_i | i \in N\}$  is the set of points belonging to  $G$ . Let

$$C_{i,j} = \begin{cases} f[x_i, \dots, x_j], & i \leq j, \\ 0, & i > j, \end{cases} \quad (1)$$

where  $f[x_i, \dots, x_j]$  denotes the divided difference of the function  $f(x)$  at points  $x_i, \dots, x_j$ , and let

$$w_i(x) = (x - x_0)(x - x_1) \cdots (x - x_{i-1}) \quad (2)$$

with  $w_0(x) = 1$ . Then  $f(x)$  can formally be expanded as the following Newton series

$$f(x) = \sum_{i=0}^{\infty} C_{0,i} w_i(x). \quad (3)$$

If  $f(x)$  is restricted on every subinterval  $[x_i, x_{i+1}]$ ,  $i = 0, 1, \dots, n-1$ , we can obtain a Newton-Hermite interpolation polynomial  $p_i(x)$  such that  $p_i(x_i) = f(x_i)$ ,  $p_i(x_{i+1}) = f(x_{i+1})$ ,  $p'_i(x_i) = f'(x_i)$  and  $p'_i(x_{i+1}) = f'(x_{i+1})$ . Obviously  $p_i(x)$  can be expressed as the following form

$$p_i(x) = a_0 + a_1(x - x_i) + a_2(x - x_i)^2 + a_3(x - x_i)^2(x - x_{i+1}), \quad (4)$$

where  $a_0 = f[x_i]$ ,  $a_1 = f[x_i, x_i]$ ,  $a_2 = f[x_i, x_i, x_{i+1}]$ ,  $a_3 = f[x_i, x_i, x_{i+1}, x_{i+1}]$  ([8]).

The counterpart of Newtons expansion is the Thiele's continued fraction expansion as follows

$$R(x) = b_0 + \frac{x - x_0}{b_1} + \frac{x - x_1}{b_2} + \cdots + \frac{x - x_{n-1}}{b_n} + \cdots, \quad (5)$$

where  $b_i = \psi[x_0, x_1, \dots, x_i]$  is the inverse difference of  $f(x)$  at  $x_0, x_1, \dots, x_i$ ,  $i = 0, 1, 2, \dots$ . If  $f(x)$  is restricted on the subinterval, a rational spline function  $R(x)$  which satisfies  $R_i(x_i) = f(x_i)$ ,  $R_i(x_{i+1}) = f(x_{i+1})$ ,  $R'_i(x_i) = f'(x_i)$ ,  $R'_i(x_{i+1}) = f'(x_{i+1})$  can be expressed as the following form

$$R_i(x) = f(x_i) + \frac{x - x_i}{b_1} + \frac{x - x_i}{b_2} + \frac{x - x_{i+1}}{b_3}, \quad (6)$$

where  $b_1 = \psi[x_i]$ ,  $b_2 = \psi[x_i, x_i]$ ,  $b_3 = \psi[x_i, x_i, x_{i+1}]$  ([8]).

### 3 The Method Based on Polynomial Splines and Rational Splines

Suppose  $f_{i-1}, f_i, f_{i+1}, f_{i+2}$  are the values of function  $f(x)$  at four equally spaced interpolation nodes. In order to improve the image detail and contour quality, the derivative is approximated by taking a weighted sum of differences of neighboring points using suitable weighting functions. In particular, we impose a low weight for large difference and higher weights for smaller intensity differences of two adjacent pixels. The proposed approximation values for derivative  $f'_i$  is

$$f'_i = \frac{G(d_{i+1})d_{i+1} + G(d_i)d_i}{G(d_{i+1}) + G(d_i)}, \quad (7)$$

where  $d_{i+1} = f_{i+1} - f_i$ ,  $d_i = f_i - f_{i-1}$ , and  $G(d)$  is a weighting function. In this paper, we set  $G(d) = 1/\sqrt{1 + d^2}$ . According to Eq. (1)–Eq. (3), the polynomial splines can be constructed as follows

$$p_i(x) = f_i + f[i, i]x + f[i, i, i + 1]x^2 + f[i, i, i + 1, i + 1]x^2(x - 1), \quad 0 \leq x \leq 1, \quad (8)$$

where

$$\begin{aligned} f[i, i] &= f'_i, \\ f[i, i, i + 1] &= f[i, i + 1] - f[i, i] = f_{i+1} - f_i - f[i, i], \\ f[i, i, i + 1, i + 1] &= f[i, i + 1, i + 1] - f[i, i, i + 1], \\ &= f[i + 1, i + 1] - f[i, i + 1] - f[i, i, i + 1], \\ &= f'_{i+1} - f_{i+1} + f_i - f[i, i, i + 1]. \end{aligned}$$

According to Eq. (6), if every  $\psi[i, i]$ ,  $\psi[i, i, i + 1]$ ,  $\psi[i, i, i + 1, i + 1]$  exist, the rational spline can be constructed as follows

$$R_i(x) = f_i + \frac{x}{\psi[i, i]} + \frac{x}{\psi[i, i, i + 1]} + \frac{x - 1}{\psi[i, i, i + 1, i + 1]}, \quad 0 \leq x \leq 1 \quad (9)$$

where

$$\begin{aligned}
 \psi[i, i] &= \frac{1}{f'_i}, \\
 \psi[i, i, i+1] &= \psi[i, i+1] - \psi[i, i] = \frac{1}{f_{i+1} - f_i} - \psi[i, i], \\
 \psi[i, i, i+1, i+1] &= \psi[i, i+1, i+1] - \psi[i, i, i+1], \\
 &= \psi[i+1, i+1] - \psi[i, i+1] - \psi[i, i, i+1], \\
 &= \frac{1}{f'_{i+1}} - \frac{1}{f_{i+1} - f_i} - \psi[i, i, i+1].
 \end{aligned}$$

Based on the above interpolation methods, the steps to resize an image can be summed up as follows.

**Step 1** Draw the four neighboring pixel as the interpolation nodes and construct the rational spline function. If the rational interpolants in some subintervals do not exist, then choose the Newton-Hermite interpolants in these subintervals instead.

**Step 2** Calculate the missing pixel between the middle two adjacent pixel using the interpolation spline.

While the pixel intensities capture the basic information about the image, divided differences and inverse differences provide the mutative information about neighboring points. As is known to all, the key characteristics of an image edge is mainly reflected by the difference between the edge data and the other data. So the above method can keep the edges since it assumes nonlinearity in the image signal and its derivatives, which is the very case in natural images.

## 4 Experimental Results

The performances of the proposed image resizing method can be deduced from some experiments. The test is conducted by down sampling the real images ( $256 \times 256$ ) by a factor of  $3/2$  and applying the proposed interpolation technique to the down sampled image ( $171 \times 171$ ). We then interpolate the down sampling image to the up sampled image ( $256 \times 256$ ), and calculate the PSNR. Table 1 shows the PSNR when we zoom the Lena image in  $3/2$  ratio size. As shown in Table 1, the nearest neighbor interpolation function, as the simplest one, is the method which has the smallest PSNR while the proposed method is the method which has the largest PSNR. The Proposed interpolator is also most efficient as shown in Fig. 1. Fig. 1 shows interpolating images of Lena using various interpolation methods. For visual comparison, the original image (Fig. 1 (a)) is down sampled to  $171 \times 171$  (Fig. 1 (b)) by a factor of  $3/2$  but enlarged to align with the  $256 \times 256$ . We can observe that more jagged contours or blurring are dramatically produced in the magnified images by the nearest neighbor and linear interpolation (Fig. 1 (c) and (d)), bicubic interpolation (Fig. 1 (e)) can improve the resizing image effect, while the proposed method can present sharper and more pleasing quality image than the conventional interpolation methods.

## 5 Conclusion

In this paper, we proposed a novel algorithm for image interpolation which can implement arbitrary image resizing. The approach is based on both the rational spline constructed by the Thiele's

Table 1: comparison of interpolation results in Lena image.

|                  | Average of error | PSNR  |
|------------------|------------------|-------|
| Nearest neighbor | 9.9761           | 21.74 |
| Linear           | 7.1373           | 25.48 |
| Cubic            | 7.0151           | 25.52 |
| Proposed method  | 5.7864           | 27.98 |



Fig. 1: Interpolated images (a) Original Lena image ( $256 \times 256$ ), (b) proposed method, (c) nearest neighbor interpolation, (d) bilinear interpolation, (e) bicubic interpolation.

interpolating continued fractions and the polynomial spline constructed by Newton-Hermite interpolation. We approximate the values of derivative using a weighted sum of differences of neighboring points with suitable weighting function. Therefore, the proposed method produces sharper and better quality images in terms of PSNR. Our approach can be applied to other interpolation applications, including numerical analysis and CAGD etc.

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